Exploring Health Patterns Using Multivariate Statistical Techniques

An Analysis of NHANES 2017–2018 Data

https://github.com/asinisusanya/nhanes-multivariate-analysis

A project report presented by:

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Abstract

This study explores health-related patterns and relationships using multivariate statistical techniques applied to the NHANES 2017–2018 dataset (n = 8,518). A structured analytical dataset was developed through careful variable selection, preprocessing, and standardization to ensure consistency and comparability across variables measured on different scales.

To gain a comprehensive understanding of the data, several multivariate methods were applied, including Principal Component Analysis (PCA), Factor Analysis (FA) with Confirmatory Factor Analysis (CFA), Discriminant Analysis (DA), Canonical Correlation Analysis (CCA), and Structural Equation Modeling (SEM). PCA reduced the dataset into five principal components explaining approximately 80.8% of the total variance, highlighting key patterns related to metabolic and cardiovascular risk. FA further revealed interpretable underlying factors such as body composition, blood pressure, and lipid profile, although the sampling adequacy was borderline (KMO = 0.498), suggesting that results should be interpreted with caution.

CCA showed a strong relationship between demographic and body-related variables and clinical health indicators, reinforcing the interconnected nature of these factors. Discriminant Analysis demonstrated how class imbalance affects classification performance, with SMOTE improving the identification of disease cases. SEM brought these findings together into a single framework, capturing how obesity, blood pressure, and metabolic variables jointly contribute to overall health risk, despite some limitations in model fit.

Overall, the results suggest that cardiovascular risk is not driven by a single factor, but by a combination of interrelated health indicators. The consistency of findings across multiple methods highlights the strength of multivariate approaches in understanding complex health data, while also emphasizing the importance of careful model validation and interpretation.

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Chapter 1

Introduction and Problem Context

1.1 Background and Motivation

The increasing availability of large-scale health data has created new opportunities for understanding population health and identifying factors that influence disease outcomes. One of the most widely used resources in this domain is the National Health and Nutrition Examination Survey (NHANES), which provides comprehensive information on demographic characteristics, clinical measurements, laboratory results, and health-related behaviors. Due to its breadth and reliability, NHANES has become a valuable source for statistical and epidemiological research.

Health outcomes, particularly those related to cardiovascular risk, are influenced by multiple interconnected factors rather than isolated variables. Characteristics such as body composition, blood pressure, lipid levels, and lifestyle behaviors interact in complex ways, making it difficult to draw meaningful conclusions using simple univariate or bivariate approaches. As a result, analyzing such data requires methods that can account for relationships among several variables simultaneously.

Previous studies have widely applied multivariate statistical techniques to health datasets such as NHANES to investigate relationships among risk factors and disease outcomes. These studies have consistently identified associations between body composition, blood pressure, and lipid profiles in relation to cardiovascular risk. However, many analyses focus on individual techniques in isolation, whereas a more integrated application of multiple multivariate methods can provide a deeper and more comprehensive understanding of the underlying data structure.

Multivariate statistical techniques provide an effective framework for addressing this complexity. These methods enable dimensionality reduction, identification of latent structures, and analysis of relationships between groups of variables. By applying a combination of such techniques to a carefully constructed subset of NHANES 2017–2018 data, this study aims to explore how multiple health indicators jointly contribute to understanding overall health risk.

1.2 Problem Definition

Despite the availability of rich and comprehensive health datasets, extracting meaningful and interpretable insights remains a significant analytical challenge. The NHANES dataset contains a large number of variables representing demographic, physiological, and clinical characteristics, many of which are interrelated and measured on different scales. This high-dimensional structure makes it difficult to identify underlying patterns and relationships using simple analytical approaches.

Traditional univariate or bivariate methods are often insufficient for capturing the joint behavior of multiple variables, as they fail to account for complex interdependencies among health indicators. Additionally, the presence of multicollinearity among variables, such as strong correlations between body composition measures or blood pressure indicators, introduces redundancy and can obscure meaningful structures within the data. Differences in measurement scales further complicate interpretation and comparison across variables.

These challenges highlight the need for a systematic analytical approach capable of simultaneously handling multiple variables, reducing dimensionality, and uncovering latent structures within the data. In particular, it is essential to identify how groups of variables interact, how underlying factors influence observed measurements, and how these relationships contribute to health outcomes.

Therefore, the core problem addressed in this study is to extract meaningful, interpretable, and statistically robust patterns from high-dimensional health data while accounting for complex interdependencies among variables. Addressing this problem requires the application of multivariate statistical techniques that can effectively model the joint behavior of multiple health indicators and provide a comprehensive understanding of their relationships.

1.3 Research Objectives

The primary objective of this study is to investigate relationships among selected health indicators using multivariate statistical techniques applied to the NHANES 2017–2018 dataset. The study aims to develop a structured and interpretable understanding of how multiple variables interact and contribute to overall health outcomes.

Specifically, the objectives of this study are:

- To construct a clean and reliable analytical dataset through appropriate variable selection and preprocessing
- To identify underlying patterns and latent structures among health-related variables using dimensionality reduction and factor-based methods
- To examine relationships between different groups of variables and understand how they interact with each other
- To apply classification techniques to assess the ability of selected variables to predict health outcomes
- To evaluate and compare the effectiveness of different multivariate techniques in capturing meaningful patterns within the data

Together, these objectives provide a comprehensive framework for analyzing complex health data and support a deeper understanding of the relationships among multiple health indicators.

1.4 Scope and Limitations

This study focuses on a selected subset of variables from the NHANES 2017–2018 dataset, representing key demographic, physiological, and clinical indicators related to cardiovascular health. The analysis is restricted to cross-sectional data and emphasizes the application of classical multivariate statistical techniques to explore patterns, relationships, and classification performance within the dataset.

Despite these strengths, several limitations should be acknowledged. The cross-sectional nature of the data prevents causal interpretation of relationships. The borderline sampling adequacy observed in factor analysis (KMO = 0.498) suggests that latent structures should be interpreted with caution. Additionally, classification results are influenced by class imbalance, and although SMOTE improves performance, it may introduce synthetic bias. Limitations in model fit within SEM further indicate that some relationships may not be fully captured by the specified model.

Overall, this chapter has established the context, defined the analytical problem, outlined the objectives, and clarified the scope of the study, providing a structured foundation for the multivariate methods applied in the subsequent analysis.

Chapter 2

Methodology

2.1 Data Description and Preparation

The analytical dataset was constructed by merging multiple NHANES 2017–2018 components using the participant identifier (SEQN), resulting in a unified dataset with 8,965 observations and 189 variables.

To reduce complexity and improve interpretability, a subset of variables was selected based on their relevance to health outcomes and suitability for multivariate analysis. The selected variables represent key dimensions of health, including demographic factors (Age, Gender, Income), body measurements (BMI, Waist), blood pressure indicators, laboratory measures (Triglycerides, Creatinine, HDL), behavioral factors (Smoking, Alcohol), and the outcome variable (Heart Disease). This selection ensures a comprehensive yet manageable representation of individual health status.

Feature engineering was applied to enhance data quality and interpretability. Multiple systolic and diastolic blood pressure readings were averaged to obtain stable measures (SYS_BP and DIA_BP), reducing measurement variability. Categorical variables originally encoded numerically were converted into meaningful labels, redundant variables were removed, and variable names were standardized for clarity.

Missing values were handled using a combination of removal and imputation. Prior to imputation, special coded values representing missing responses (e.g., 7 and 9) were converted to standard missing values (NaN). Observations with missing values in the outcome variable were removed to ensure reliability in subsequent analysis. For the remaining variables, numerical variables were imputed using median values, while categorical variables were imputed using the mode. This approach ensures robustness to outliers and preserves dominant data patterns, resulting in a complete dataset with no missing values.

Outliers were identified using both visual methods and the Z-score approach. Instead of removing extreme values, transformations were applied to reduce their influence. Specifically, log transformations were applied to skewed variables such as Income and Triglycerides to improve distributional balance.Finally, all numerical variables were standardized to have mean approximately zero and unit variance. This step ensures comparability across variables measured on different scales.

The final dataset consists of 8,518 observations and 14 variables, forming a clean and well-structured dataset suitable for subsequent multivariate analysis.

2.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to examine the structure and characteristics of the dataset prior to applying multivariate techniques. Variables were first classified into numerical and categorical types to ensure appropriate analytical methods.

Descriptive statistics, including measures of central tendency, variability, skewness, and kurtosis, were computed for numerical variables to understand their distributional properties. Histograms with density

curves were used to further assess the shape and spread of the variables.

To investigate relationships among variables, a correlation matrix was computed and visualized using a heatmap. This step helps identify dependencies and assess potential multicollinearity.In addition, scatter plots were generated for selected variable pairs to explore pairwise relationships and assess linearity. A pairplot (scatter matrix) was also used to examine interactions among multiple variables simultaneously.

These analyses provide a foundational understanding of the dataset and support the suitability of the data for subsequent multivariate analysis.

2.3 Multivariate Analysis Framework

Following EDA, a structured multivariate analysis framework was adopted to systematically investigate different aspects of the dataset. The analysis was designed to move from data reduction and structure identification to classification and relationship modeling.

Principal Component Analysis (PCA) was first applied to reduce dimensionality and identify dominant patterns among correlated variables. This was followed by Factor Analysis (FA) and Confirmatory Factor Analysis (CFA) to examine underlying latent structures and validate relationships among variables.

To address classification objectives, Discriminant Analysis (DA) was employed using both Linear and Quadratic approaches. During this stage, class imbalance in the outcome variable was identified, and the Synthetic Minority Over-sampling Technique (SMOTE) was applied to improve class representation and enhance classification performance.

Canonical Correlation Analysis (CCA) was then used to explore relationships between groups of variables, providing insight into interdependencies across different dimensions of the dataset. Finally, Structural Equation Modeling (SEM) was applied to model complex relationships among observed and latent variables within a unified framework.

This sequential approach ensures a comprehensive analysis of the dataset, where each method contributes to understanding different structural and relational aspects of the data. The overall analytical workflow adopted in this study is illustrated in Figure 2.1.

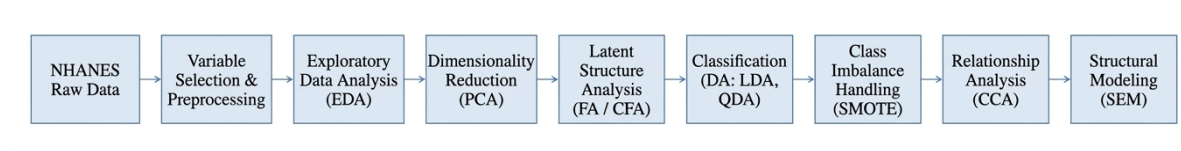


Figure 2.1: Overview of the multivariate analysis framework

2.3.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset and to identify dominant patterns among correlated variables.

The analysis was conducted using numerical variables, including Age, Income, BMI, Waist, Triglycerides, Creatinine, HDL, SYS_BP, and DIA_BP. These variables represent key demographic, physiological, and clinical measurements relevant to health outcomes.

The k-th principal component is defined as:

$$Z_k = a_k^T X$$

where a_k is a vector of coefficients chosen to maximize the variance of the component.

PCA assumes that variables are continuous, exhibit linear relationships, and are sufficiently correlated. Additionally, the data is required to be standardized to ensure comparability across variables.These assumptions are satisfied in this study, as the selected variables are continuous health-related measurements, prior analysis confirmed the presence of meaningful correlations, and standardization was applied during data preprocessing.

PCA is appropriate in this context due to the presence of correlated variables (e.g., BMI and Waist, SYS_BP and DIA_BP), making it suitable for reducing redundancy while preserving the most important information in the dataset.

2.3.2 Factor Analysis (FA)

Factor Analysis (FA) was applied to identify underlying latent structures among the observed variables and to reduce dimensionality while maintaining interpretability.

The analysis was conducted using numerical variables including Age, Income, BMI, Waist, Triglycerides, Creatinine, HDL, SYS_BP, and DIA_BP, which represent key aspects of individual health and are expected to be influenced by common underlying factors.

The factor model is given by:

$$X = \Lambda F + \epsilon$$

where Λ is the factor loading matrix, F represents latent factors, and ϵ denotes unique error terms.

To assess the suitability of the data for factor analysis, Bartlett's Test of Sphericity and the Kaiser-Meyer-Olkin (KMO) measure were computed. Bartlett's test resulted in a statistically significant p-value (p ¡ 0.05), indicating that the correlation matrix is not an identity matrix and that sufficient correlations exist among variables. The KMO measure yielded a value of 0.498, indicating insufficient sampling adequacy, which suggests limitations in the suitability of the data for factor analysis. FA assumes linear relationships among variables, sufficient correlation, and approximate multivariate normality. Based on these diagnostic tests and preprocessing steps, the data was considered acceptable for applying factor analysis.

FA is appropriate in this context as health-related variables are often influenced by latent constructs such as metabolic and cardiovascular factors, making it useful for uncovering interpretable underlying structures.

Confirmatory Factor Analysis (CFA) was subsequently applied to validate the hypothesized factor structure. In CFA, the relationships between observed variables and latent factors are specified in advance, and model adequacy is assessed using goodness-of-fit measures. This step ensures that the identified factor structure is consistent with the observed data.

2.3.3 Discriminant Analysis (DA)

Discriminant Analysis (DA) was applied to classify individuals based on health-related variables, with the objective of predicting the presence of heart disease.

The analysis used numerical variables including Age, Income, BMI, Waist, Triglycerides, Creatinine, HDL, SYS_BP, and DIA_BP as predictors, while Heart_Disease was used as the target variable.

Initial examination of the dataset revealed a class imbalance problem in the target variable, where one class was underrepresented. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to generate synthetic samples for the minority class, improving class balance and supporting more reliable classification.

Both Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) were implemented. LDA assumes multivariate normality and equal covariance matrices across classes, while QDA relaxes the equal covariance assumption and allows for class-specific covariance structures.

Several assumption checks were conducted to evaluate the suitability of the data. Box's M test indicated that the covariance matrices are not equal across groups (p ¡ 0.05), suggesting a violation of the equal

covariance assumption required for LDA. The Shapiro-Wilk test indicated deviations from normality (p ¡ 0.05) across variables. Correlation analysis revealed multicollinearity between certain variables (e.g., BMI and Waist), while group mean comparisons indicated meaningful differences between classes.

Despite these violations, the application of LDA remains justified. In large real-world datasets, strict adherence to assumptions such as normality and equal covariance is often unrealistic. LDA is known to be relatively robust to moderate violations of these assumptions, particularly when sample sizes are sufficiently large. Additionally, LDA provides a simple and interpretable baseline model, allowing comparison with more flexible approaches such as QDA, which explicitly accounts for unequal covariance structures.

Discriminant Analysis is appropriate in this context as the objective involves classification into predefined groups. The combined use of LDA, QDA, and SMOTE enables a comprehensive evaluation of classification performance under different modeling assumptions and data conditions.

2.3.4 Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis (CCA) was applied to examine the relationships between two sets of variables representing different aspects of health.

The analysis was conducted by grouping variables into two sets. The first set included demographic and body-related variables such as Age, Income, BMI, and Waist, while the second set consisted of clinical and laboratory variables including Triglycerides, Creatinine, HDL, SYS_BP, and DIA_BP. This grouping allows the investigation of relationships between lifestyle/body characteristics and physiological health indicators.

CCA seeks to identify linear combinations of variables from each set such that the correlation between these combinations is maximized. Formally, it determines vectors a and b such that:

$$\rho = corr(a^T X,\, b^T Y)$$

CCA assumes linear relationships between variables, absence of severe multicollinearity within each set, and approximate multivariate normality. These assumptions are considered reasonable given the preprocessing steps and the nature of the selected variables.

CCA is appropriate in this context as it enables the simultaneous analysis of relationships between two groups of variables, providing insights into how demographic and body-related factors are associated with clinical health indicators.

2.3.5 Structural Equation Modeling (SEM)

Structural Equation Modeling (SEM) was applied to model complex relationships among observed variables and latent constructs within a unified framework.

The analysis included observed variables such as Age, Income, BMI, Waist, Triglycerides, Creatinine, HDL, SYS_BP, and DIA_BP, which were used to represent underlying latent constructs related to demographic, physiological, and clinical factors.

SEM consists of two components: the measurement model, which relates observed variables to latent variables, and the structural model, which specifies relationships among latent variables. The models can be expressed as:

$$X = \Lambda F + \epsilon, \quad F = BF + \zeta$$

where Λ represents factor loadings, B denotes relationships among latent variables, and ϵ and ζ are error terms.

SEM assumes linear relationships, correct model specification, and approximate multivariate normality. These assumptions are considered reasonable given the preprocessing steps and the use of continuous health-related variables.

SEM is appropriate as it models multiple relationships simultaneously while incorporating latent constructs, providing a comprehensive framework for understanding interconnected health factors.

Chapter 3

Results and Interpretation

3.1 Exploratory Analysis Results

The dataset consists of 8,518 observations and 14 variables, with nine numerical variables used for analysis. All numerical variables were standardized (mean = 0, std = 1).

Most variables follow approximately normal distributions after transformation, although BMI and HDL show slight right skewness, while Creatinine remains highly skewed with extreme values, indicating variability in certain clinical measurements.

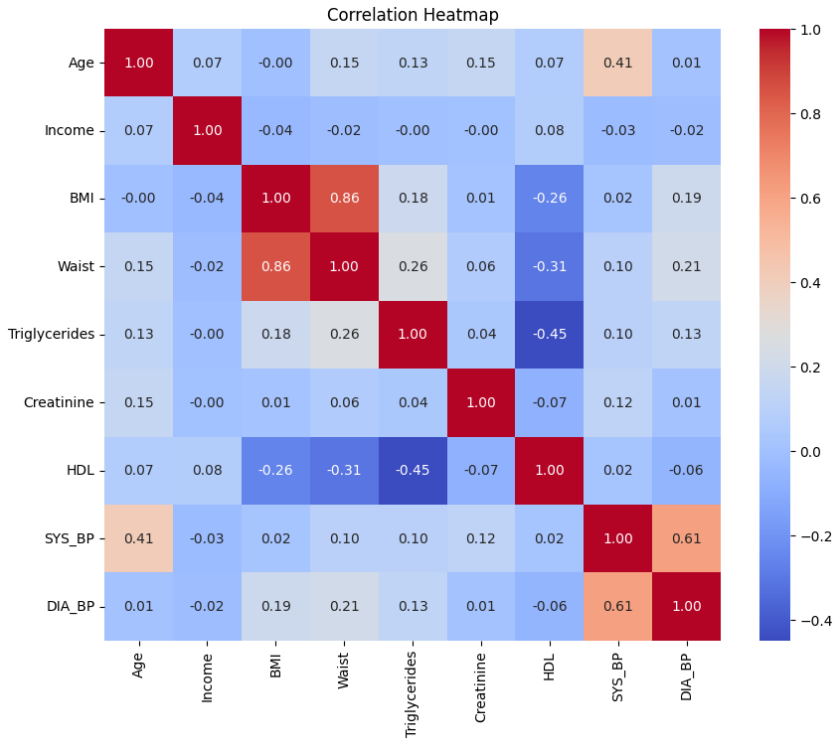


Figure 3.1: Correlation matrix of numerical variables

Correlation analysis reveals several meaningful relationships. Strong positive correlations are observed between BMI and Waist (0.86), and between SYS_BP and DIA_BP (0.61). Moderate relationships are identified between Age and SYS_BP (0.41), and between Waist and Triglycerides (0.26). Negative correlations are observed between Triglycerides and HDL (0.45), as well as between body fat measures and HDL. Income shows weak relationships with most variables.

Further analysis of pairwise relationships indicates a moderate positive trend between Age and systolic blood pressure, a weak relationship between BMI and systolic blood pressure, and a clear inverse relation-

ship between Triglycerides and HDL.

Overall, the dataset exhibits meaningful patterns consistent with known health relationships. Despite some skewness and outliers, particularly in clinical variables, the data is well-structured and suitable for subsequent multivariate analysis.

3.2 Principal Component Analysis Results

Principal Component Analysis (PCA) was applied to nine standardized numerical variables to reduce dimensionality and identify underlying patterns in the data.

The selection of principal components was based on multiple criteria. The Kaiser criterion suggests retaining components with eigenvalues greater than one, which corresponds to the first four components, while the fifth component is borderline. The scree plot shows a clear elbow around the fourth to fifth component, indicating diminishing returns beyond this point. The cumulative explained indicates that the first four components explain 69.8% of the total variance, while the first five components explain 80.8%. Based on these criteria, five principal components were retained.

The first principal component (PC1) explains 26.4% of the variance and is primarily associated with BMI, Waist, Triglycerides, and blood pressure variables, with an inverse contribution from HDL. This represents a metabolic and cardiovascular risk dimension. The second component (PC2), explaining 18.3%, is dominated by Age and blood pressure variables, indicating a blood pressure–aging relationship. The third component (PC3), explaining 12.8%, highlights the relationship between Triglycerides, Creatinine, and HDL, suggesting a metabolic–renal interaction. The fourth component (PC4), explaining 12.3%, is influenced by Income, Age, and HDL, indicating a socioeconomic-health dimension. The fifth component (PC5), explaining 11.0%, is primarily driven by Income, representing a relatively independent socioeconomic dimension.

The PCA biplot (Figure 3.2) provides a visual representation of these relationships. BMI, Waist, and Triglycerides cluster together, confirming strong positive associations, while HDL points in the opposite direction, indicating an inverse relationship with metabolic risk factors. SYS_BP and DIA_BP are closely aligned, reflecting strong correlation, and Age aligns with blood pressure variables, supporting the interpretation of PC2.

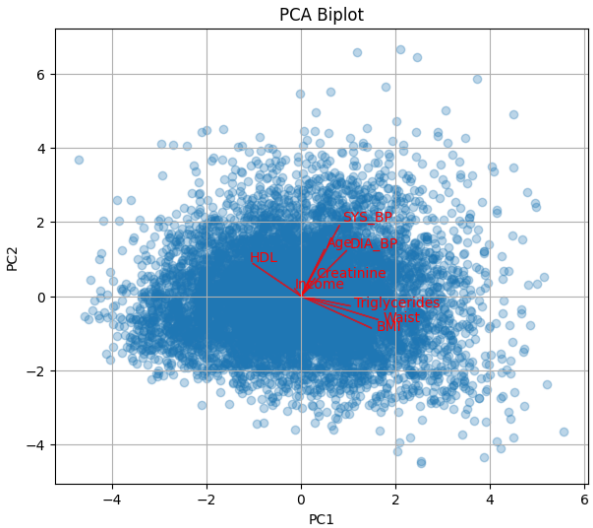


Figure 3.2: PCA biplot showing relationships among variables

Overall, PCA effectively reduces the original nine variables into five interpretable components while preserving a substantial proportion of the total variability. The identified components correspond to meaningful health-related dimensions, demonstrating that the dataset captures realistic physiological and socioeconomic patterns.

3.3 Factor Analysis and Confirmatory Factor Analysis Results

Factor Analysis (FA) was applied to nine standardized numerical variables to identify underlying latent structures. Bartlett's test was highly significant (p ¡ 0.05), indicating sufficient correlations among variables, while the KMO measure (0.498) suggests borderline sampling adequacy.

Factor selection was based on eigenvalues and the scree plot . The first four factors have eigenvalues greater than one, and the scree plot shows an elbow around the fourth factor, indicating diminishing contribution beyond this point. Accordingly, four factors were retained for interpretation.
The extracted factors reveal meaningful latent dimensions. The first factor is strongly associated with BMI and Waist, representing a body composition (obesity) dimension. The second factor is dominated by SYS_BP and DIA_BP, capturing a blood pressure dimension. The third factor is primarily driven by HDL with an inverse contribution from triglycerides, reflecting a lipid profile dimension. The fourth factor is dominated by Age, representing a demographic dimension.

Communality values indicate that most physiological variables are well explained (e.g., Waist, HDL, Age), while variables such as Income and Creatinine have low communalities, suggesting weaker alignment with the latent structure. The four factors together explain 59.4% of the total variance, which is reasonable given that factor analysis focuses on shared variance.

Confirmatory Factor Analysis (CFA) was conducted to validate a three-factor model based on theoretical constructs: Obesity (BMI, Waist), Blood Pressure (SYS_BP, DIA_BP), and Metabolic (HDL, Triglycerides). Factor loadings indicate strong relationships within each construct, with BMI and Waist defining the obesity factor, SYS_BP and DIA_BP defining the blood pressure factor, and HDL and triglycerides capturing the metabolic dimension with an inverse relationship.

Correlations among latent factors suggest that these constructs are related but distinct. However, diagnostic outputs indicate some model limitations. The presence of negative variance estimates (Heywood case) and high modification indices suggests potential model misspecification and unaccounted relationships between variables, particularly between body composition and blood pressure measures.

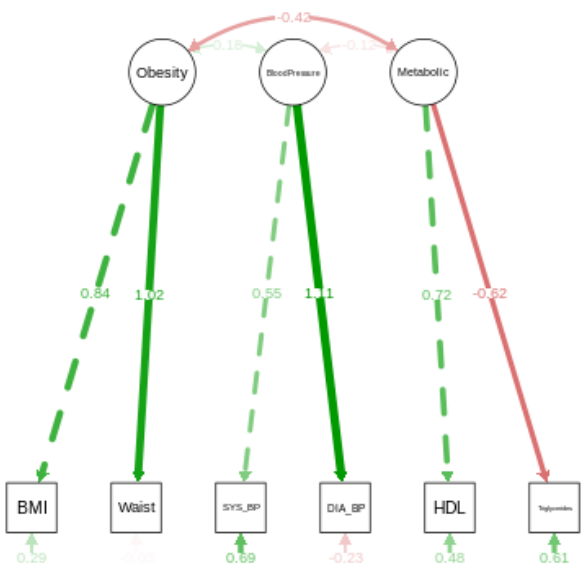


Figure 3.3: CFA model illustrating relationships between latent variables and observed indicators

Overall, FA identifies clear latent health dimensions, while CFA provides partial confirmation of the hypothesized structure. Although the model captures meaningful relationships consistent with theoretical expectations, the presence of estimation issues suggests that further refinement may improve model fit.

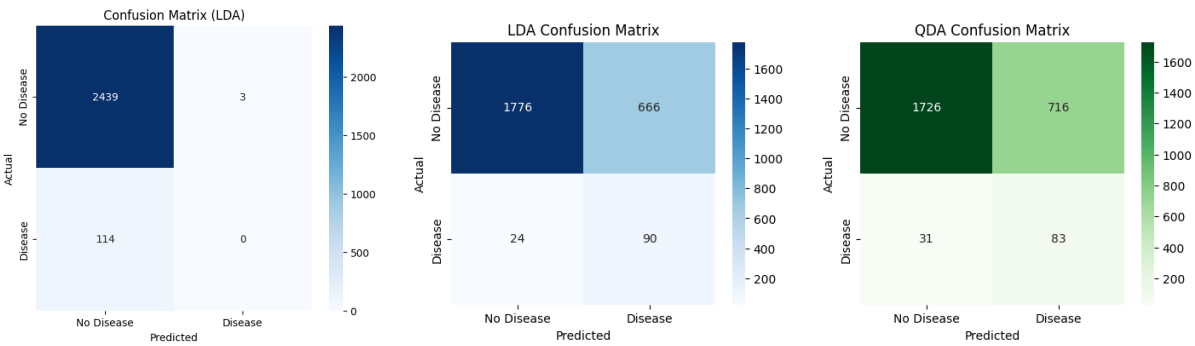
3.4 Discriminant Analysis Results

Discriminant Analysis was applied to classify individuals based on health-related variables, with the objective of predicting the presence of heart disease.

The LDA model indicates that variables such as Age, Waist, and SYS_BP contribute positively to disease classification, while BMI, HDL, and DIA_BP show negative contributions. This suggests that higher age, central obesity, and elevated systolic blood pressure are associated with increased disease risk, while higher HDL and lower body mass are associated with reduced risk.

Classification performance across models is summarized in Figure 3.4. The LDA model applied to the original dataset shows strong bias toward the majority class, correctly classifying most "No Disease" cases but failing to identify "Disease" cases effectively. This highlights the impact of class imbalance.

After applying SMOTE, the LDA model demonstrates improved detection of the minority class, resulting in more balanced classification. The QDA model, applied to the SMOTE-balanced dataset, provides comparable performance while allowing for more flexible class boundaries.



(a) LDA (Original) (b) LDA + SMOTE (c) QDA + SMOTE

Figure 3.4: Comparison of classification performance across models

Overall, the results indicate that class imbalance significantly affects classification performance. The application of SMOTE improves the detection of disease cases, while both LDA and QDA provide useful but imperfect classification. The findings suggest that variables such as age, body composition, and blood pressure are important predictors, although some overlap between classes limits classification accuracy.

3.5 Canonical Correlation Analysis Results

Canonical Correlation Analysis (CCA) was applied to examine the relationship between two sets of variables: demographic/body-related variables (Age, Income, BMI, Waist) and clinical variables (Triglycerides, Creatinine, HDL, SYS_BP, DIA_BP).

The first canonical correlation indicates a meaningful association between the two variable sets, suggesting that variations in body composition and demographic factors are related to clinical health indicators.Subsequent canonical correlations were relatively smaller and contributed less to explaining the relationship between the two variable sets. Therefore, interpretation focused primarily on the first canonical function.

Examination of canonical loadings shows that BMI and Waist contribute strongly to the first set, while HDL, Triglycerides, and blood pressure variables contribute significantly to the second set. This indicates

that body fat measures are closely linked to lipid profiles and cardiovascular indicators.

The relationship suggests that individuals with higher body mass and central adiposity tend to exhibit unfavorable lipid profiles and elevated blood pressure, reflecting an underlying metabolic risk pattern. Income shows relatively weak contribution, indicating limited direct influence in this multivariate relationship.

Overall, the CCA results highlight a strong connection between body composition and clinical health indicators, supporting findings from PCA and Factor Analysis and confirming the presence of meaningful multivariate relationships within the dataset.

3.6 Structural Equation Modeling Results

Structural Equation Modeling (SEM) was applied to examine relationships among latent health-related constructs, extending the factor structure identified in previous analyses.

The model specifies a latent variable representing overall health risk, measured using observed variables including BMI, Waist, Triglycerides, HDL, SYS_BP, and DIA_BP. Additional observed variables such as smoking, alcohol consumption, and income were included to account for behavioral and socioeconomic influences.

Factor loadings indicate that BMI and Waist contribute strongly to the latent construct, reflecting body composition effects. Blood pressure variables (SYS_BP and DIA_BP) also show strong positive contributions, while HDL exhibits a negative relationship, consistent with its protective role. Triglycerides show a moderate positive contribution, supporting their role in metabolic risk.

The SEM diagram illustrates these relationships, where most observed variables positively contribute to the latent health risk factor, while HDL acts in the opposite direction.

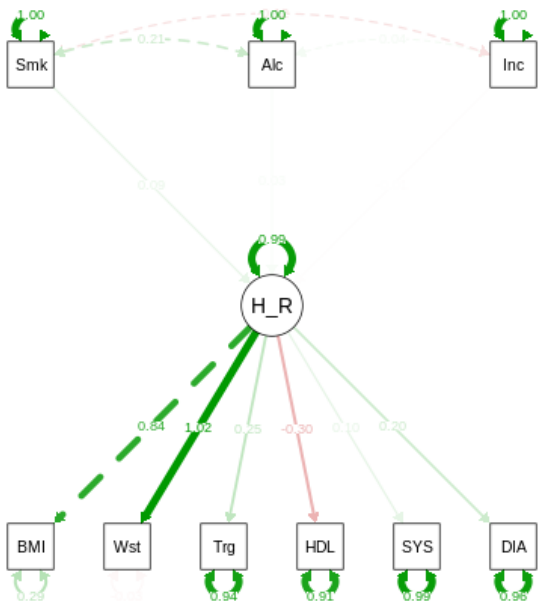


Figure 3.5: Structural Equation Model illustrating relationships between observed variables and latent health risk

The results suggest that the latent construct captures a combination of obesity, blood pressure, and metabolic factors, consistent with findings from PCA and Factor Analysis. Behavioral variables show

relatively weaker contributions, indicating that physiological variables dominate the latent structure.

However, model diagnostics indicate some limitations. The presence of weak paths and variability in contributions suggests that the model may not fully capture all relationships among variables. This may be due to model simplification or unaccounted interactions between predictors.

Overall, SEM provides a comprehensive framework for integrating multiple health indicators into a single latent construct. While the model successfully captures key relationships, further refinement such as incorporating additional paths or restructuring latent variables may improve model fit and explanatory power.

3.7 Method Comparison

The applied multivariate techniques provide complementary insights into the dataset, each addressing different analytical objectives.

PCA reduced the nine variables into five principal components explaining 80.8% of the total variance, effectively capturing major patterns such as metabolic risk and blood pressure–age relationships. In contrast, Factor Analysis extracted four factors explaining 59.4% of the variance, identifying interpretable latent constructs such as obesity, blood pressure, lipid profile, and age. While PCA achieved stronger dimensionality reduction, FA provided more meaningful interpretation of underlying health-related structures, although its effectiveness was limited by borderline sampling adequacy.

Discriminant Analysis highlighted challenges in classification due to class imbalance. The LDA model applied to the original dataset showed strong bias toward the majority class and failed to identify disease cases effectively. After applying SMOTE, classification performance improved significantly, with more balanced detection of both classes. The QDA model provided comparable performance while allowing more flexible class boundaries, indicating that relaxing covariance assumptions can improve model adaptability. However, some overlap between classes remained, limiting overall classification accuracy.

CCA revealed a meaningful relationship between body composition variables (BMI, Waist) and clinical indicators (HDL, Triglycerides, blood pressure), confirming that individuals with higher adiposity tend to exhibit adverse metabolic and cardiovascular profiles. This relationship is consistent with patterns identified in PCA and FA, reinforcing the presence of strong multivariate associations.

SEM integrated multiple variables into a unified framework, capturing a latent health risk construct influenced by obesity, blood pressure, and metabolic factors. While the model aligned well with theoretical expectations, diagnostic results indicated limitations such as estimation issues and potential model misspecification, suggesting that further refinement is needed to improve model fit.

Overall, PCA was most effective for dimensionality reduction, FA provided clearer interpretation of latent structures, DA methods enabled classification (significantly improved by SMOTE), CCA identified relationships between variable groups, and SEM offered a comprehensive but more complex modeling framework. Together, these methods provide a holistic understanding of the dataset.

Chapter 4

Discussion

The results of the multivariate analyses reveal several consistent and meaningful health-related patterns within the dataset. Across multiple techniques, body composition, lipid profile, and blood pressure emerge as key factors associated with cardiovascular risk.

PCA identified a dominant component representing metabolic and cardiovascular risk, characterized by strong contributions from BMI, Waist, Triglycerides, and blood pressure variables, with an inverse relationship with HDL. This pattern was reinforced by Factor Analysis, which extracted a clear obesity factor and a blood pressure factor, indicating that these variables form distinct but related dimensions of health. Similarly, CCA demonstrated a strong relationship between body composition variables and clinical indicators, confirming that individuals with higher adiposity tend to exhibit adverse lipid profiles and elevated blood pressure.

SEM further integrated these findings by modeling a latent health risk construct influenced by obesity, blood pressure, and metabolic factors. Although the model showed some limitations, including estimation issues and potential misspecification, it successfully captured the interconnected nature of these health indicators. Together, these results highlight that cardiovascular risk is not driven by a single variable but by a combination of interrelated physiological factors.

The classification results from Discriminant Analysis provide additional insight into the practical challenges of disease prediction. The poor performance of LDA on the original dataset highlights the impact of class imbalance, where the model failed to identify disease cases effectively. The application of SMOTE significantly improved classification performance, emphasizing the importance of data balancing in real-world predictive modeling. However, the observed overlap between classes suggests that health conditions such as heart disease are complex and may not be easily separable using linear or even non-linear boundaries.

A key strength of this study is the consistency of findings across multiple methods. PCA, FA, CCA, and SEM all indicate strong relationships between obesity, lipid imbalance, and blood pressure, reinforcing the reliability of the identified patterns. These findings align with established medical knowledge, where increased body fat, abnormal lipid levels, and elevated blood pressure are known risk factors for cardiovascular disease.

From a real-world perspective, the results highlight the importance of considering multiple health indicators simultaneously rather than in isolation. The strong interdependence among variables suggests that effective prevention and intervention strategies should target combinations of risk factors, such as improving body composition, managing lipid levels, and controlling blood pressure. Additionally, the classification results demonstrate the challenges of early disease detection and the need for balanced and well-structured datasets in predictive health models.

Overall, the analysis demonstrates that multivariate techniques provide a comprehensive framework for understanding complex health relationships. By combining multiple methods, it is possible to uncover consistent patterns, validate findings, and gain deeper insight into the underlying structure of the data.

Chapter 5

Conclusion and Future Work

This study applied multiple multivariate statistical techniques to analyze the NHANES dataset and investigate relationships among demographic, physiological, and clinical variables related to cardiovascular health. Methods including PCA, FA, DA, CCA, and SEM were used to explore data structure, identify latent patterns, model relationships, and perform classification.

The results reveal consistent health-related patterns across methods. Body composition, lipid profile, and blood pressure emerged as key factors associated with cardiovascular risk. PCA and FA identified meaningful underlying structures, while CCA confirmed strong relationships between body composition and clinical indicators. DA highlighted the impact of class imbalance on classification performance, with SMOTE significantly improving the detection of disease cases. SEM further integrated these findings into a unified framework, capturing the combined influence of multiple risk factors.

From a methodological perspective, each technique provided complementary insights. PCA was effective for dimensionality reduction, FA improved interpretability, DA enabled classification, CCA revealed inter-set relationships, and SEM offered an integrated modeling approach. Together, these methods contributed to a comprehensive understanding of the dataset.

However, several limitations were identified. The presence of class imbalance affected classification performance, requiring the use of resampling techniques. FA showed borderline sampling adequacy, and SEM results indicated potential model misspecification and estimation issues. Additionally, some variables exhibited skewness and outliers, which may influence model assumptions.

Future work can focus on improving model performance and robustness. This includes refining the SEM structure, incorporating additional relevant variables, and applying advanced machine learning techniques for classification. Further studies may also explore longitudinal data or alternative modeling approaches to better capture complex health relationships.

Overall, this study demonstrates the value of multivariate techniques in uncovering meaningful patterns in health data and highlights the importance of integrating multiple analytical approaches to gain deeper insight into complex systems.

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Appendix A

Dataset

The dataset used in this study was obtained from the National Health and Nutrition Examination Survey (NHANES) 2017–2018 cycle, provided by the Centers for Disease Control and Prevention (CDC). The dataset is publicly available at:

https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2017

For this analysis, multiple NHANES data components were utilized. Specifically, eight XPT data files were extracted and merged using the unique participant identifier (SEQN). These files include demographic, examination, and laboratory data relevant to the selected variables. The raw data files used in this study are available in the project repository:

Raw NHANES Data Files (XPT Format)

Due to the large size and high dimensionality of the NHANES dataset, only a subset of relevant variables was selected for analysis. These include Age, Income, BMI, Waist, Triglycerides, Creatinine, HDL, SYS BP, and DIA BP.
After preprocessing steps including cleaning, transformation, and standardization, the final analytical dataset was generated. The processed dataset used in this study is available at:

Final Processed Dataset (CSV)

A sample of the processed (standardized) dataset used in the analysis is presented below.

Table 1: Sample of processed dataset

Age	Income	BMI	Waist	Triglycerides	Creatinine	HDL	SYS_BP	DIA_BP
0.44	-0.12	1.03	1.02	-0.58	0.12	0.16	-1.39	-1.84
-0.93	-0.88	-1.08	-0.84	0.79	0.09	-0.74	-0.59	-0.66
1.17	-0.17	0.90	0.52	0.48	-0.02	-1.58	0.51	-0.42
0.48	0.99	1.22	0.28	0.81	-0.61	-0.65	0.80	-0.18
...	...	...	...	...	...	...	...	...

Appendix B

B.1 Code

The complete implementation used for data preprocessing, analysis, and modeling is available at:

Full Analysis Notebook (Jupyter)

The repository includes all scripts used for exploratory data analysis (EDA), Principal Component Analysis (PCA), Factor Analysis (FA), Confirmatory Factor Analysis (CFA), Discriminant Analysis (LDA and QDA with SMOTE), Canonical Correlation Analysis (CCA), and Structural Equation Modeling (SEM).

In addition to the code, the repository contains detailed explanations and extended interpretations for each analytical step. These provide a more comprehensive discussion of the results, complementing the concise interpretations presented in the main report.

A visual overview of the notebook structure is shown below.

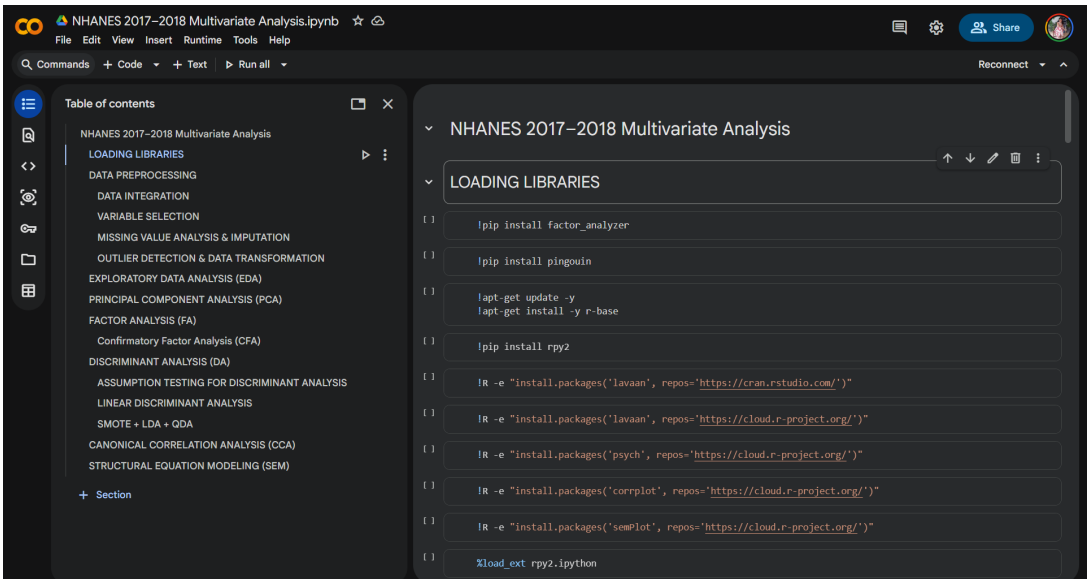


Figure 1: Structure of the Jupyter Notebook used for the analysis

B.2 Analytical Pipelines (Pseudocode)

Data Preparation Pipeline

```
LOAD NHANES XPT files
MERGE datasets using SEQN
SELECT relevant variables
HANDLE missing values:
    - Remove records with missing target
    - Impute numerical variables (median)
    - Impute categorical variables (mode)
APPLY log transformation to skewed variables
STANDARDIZE numerical variables
OUTPUT cleaned dataset
```

Exploratory Data Analysis (EDA)

```
COMPUTE descriptive statistics
PLOT distributions (histograms)
COMPUTE correlation matrix
VISUALIZE correlation heatmap
GENERATE scatter plots and pairplots
IDENTIFY patterns and relationships
```

Principal Component Analysis (PCA)

```
INPUT standardized numerical variables
COMPUTE covariance matrix
EXTRACT eigenvalues and eigenvectors
SELECT components using:
    - Eigenvalue > 1 (Kaiser criterion)
    - Scree plot
PROJECT data onto principal components
INTERPRET component loadings
```

Factor Analysis (FA & CFA)

```
CHECK suitability (KMO, Bartlett test)
EXTRACT factors using eigenvalues
ROTATE factors for interpretability
INTERPRET factor loadings
SPECIFY CFA model
ESTIMATE model parameters
EVALUATE model fit
```

Discriminant Analysis (LDA & QDA)

```
INPUT predictor variables and target variable
CHECK class distribution
APPLY SMOTE to balance classes
TRAIN LDA model
TRAIN QDA model
EVALUATE classification performance
COMPARE model results
```

Canonical Correlation Analysis (CCA)

```
DEFINE two variable sets:
    - Set 1: demographic/body variables
    - Set 2: clinical variables
COMPUTE canonical variates
MAXIMIZE correlation between sets
ANALYZE canonical loadings
INTERPRET relationships between variable groups
```

Structural Equation Modeling (SEM)

```
DEFINE latent variables
SPECIFY measurement model
SPECIFY structural model
ESTIMATE parameters
EVALUATE model fit
INTERPRET path coefficients
```

Appendix C

Additional Figures

All figures generated during the analysis are available in the project repository:

Complete Set of Figures (GitHub Repository)

Additional visualizations are provided below to support the findings presented in the main report. Only selected figures that are essential for understanding the analysis are included here to maintain clarity and avoid redundancy.

C.1 Exploratory Data Analysis (EDA)

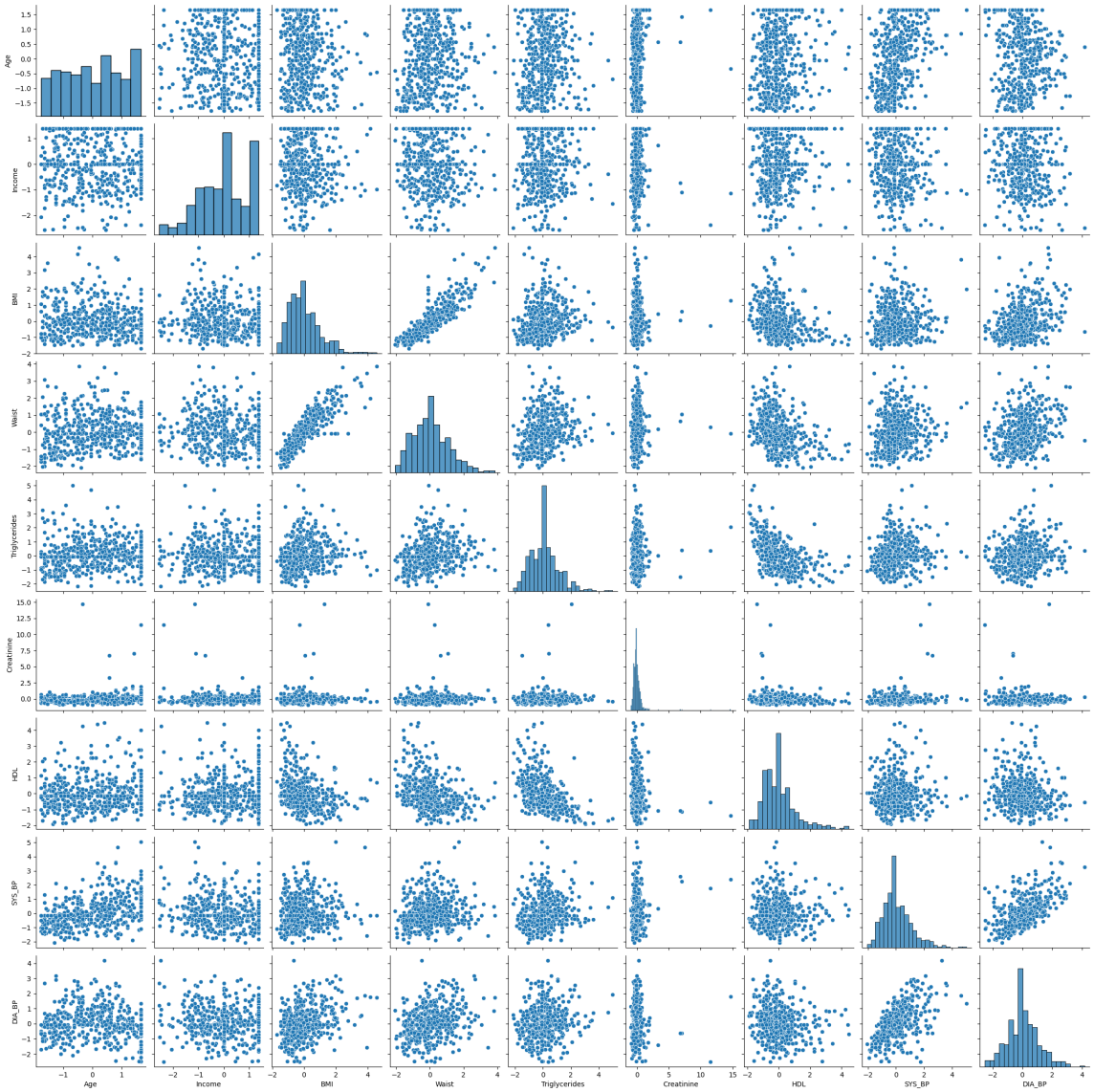
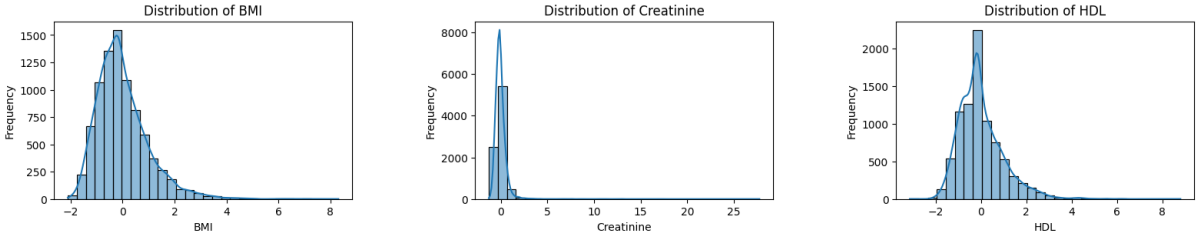
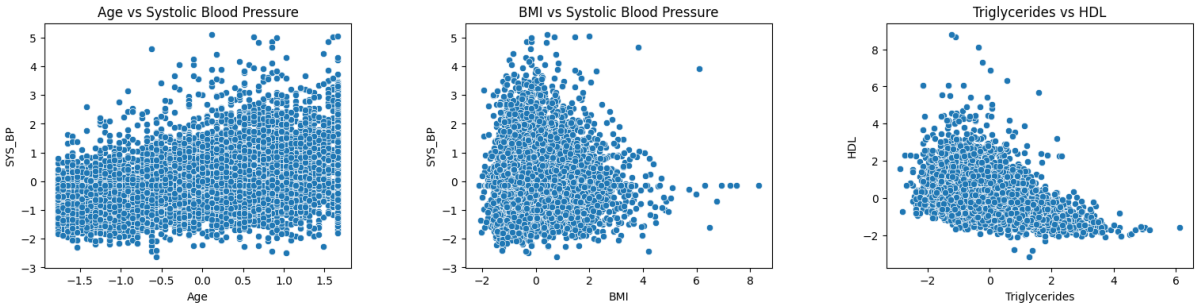


Figure 2: Pairwise relationships and distributions of selected variables, illustrating correlations, clustering patterns, and variability across the dataset



(a) BMI (b) Creatinine (c) HDL

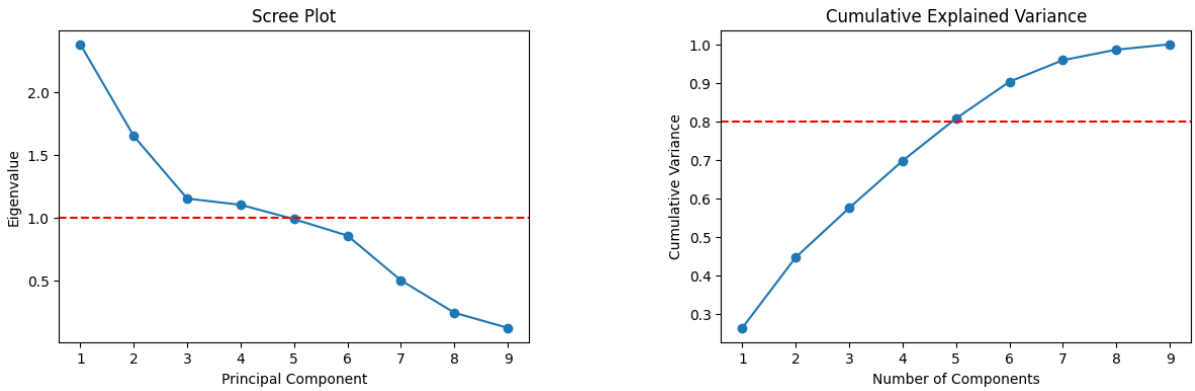
Figure 3: Distribution of selected variables illustrating variability and skewness



(a) Age vs SYS_BP (b) BMI vs SYS_BP (c) Triglycerides vs HDL

Figure 4: Key relationships between selected variables

C.2 Principal Component Analysis (PCA)



(a) Scree plot (b) Cumulative explained variance

Figure 5: Component selection criteria for PCA

C.3 Factor Analysis (FA)

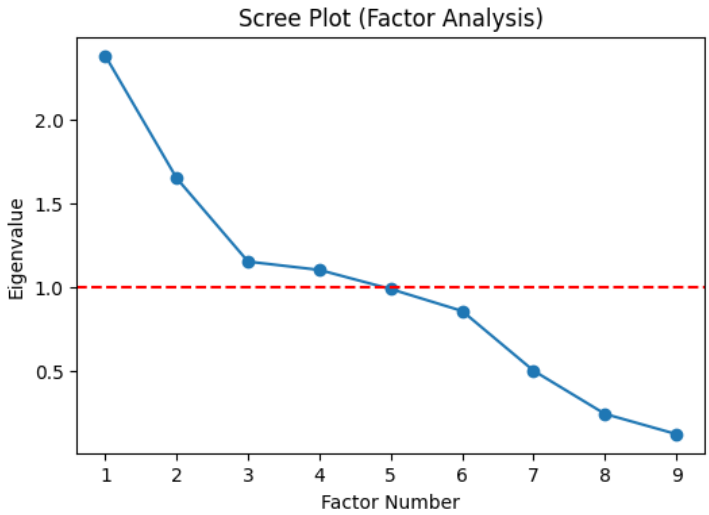
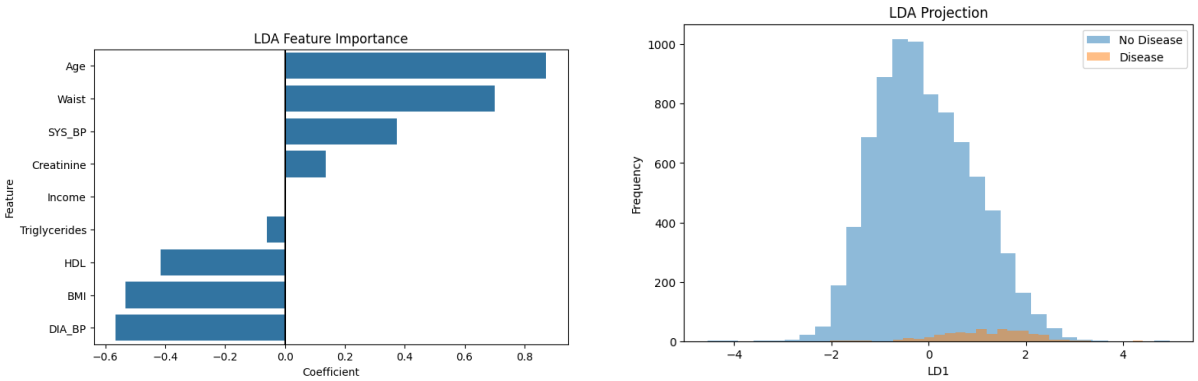


Figure 6: Scree plot for factor selection

C.4 Discriminant Analysis (DA)



(a) LDA Feature Importance

(b) LDA Projection (LD1)

Figure 7: Additional Discriminant Analysis results showing variable contributions and class separation